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Process and Measurement Noise Estimation for Kalman Filtering

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ABSTRACT

The Kalman filter gain can be extracted from output signals but the covariance of the state error cannot be evaluated without knowledge of the covariance of the process and measurement noise Q and R. Among the methods that have been developed to estimate Q and R from measurements the two that have received most attention are based on linear relations between these matrices and: 1) the covariance function of the innovations from any stable filter or 2) the covariance function of the output measurements. This paper reviews the two approaches and offers some observations regarding how the initial estimate of the gain in the innovations approach may affect accuracy.

Keywords: Kalman Filter, Process Noise, Measurement Noise

NOMENCLATURE

A	Discrete system matrix
B	Input matrix of dynamic system
C	Measurement matrix of state space model
E	Expected value operator
K	Steady state Kalman gain
K_0	Initial steady state Kalman gain
$\hat{l}_j$	Estimate of the covariance function of the innovation process
l_j	Covariance function of the innovation process
Ob	Observability matrix
P_0	Initial state error covariance matrix
P	State error covariance matrix
x_0	Initial state vector
x_k	State vector at time k
$\hat{x}_k^-$	State estimate after the measurement at time k is taken into consideration
$\hat{x}_k^-$	State estimate before the measurement at time k is taken into consideration

y_k	Discrete measurement vector
w_k	Discrete process noise
v_k	Discrete measurement noise
$\hat{R}$	Estimate of the covariance matrix of the measurement noise
R	Covariance matrix of the measurement noise
$\hat{Q}$	Estimate of the covariance matrix of the process noise
Q	Covariance matrix of the process noise
Σ	State covariance matrix
Λ_j	Covariance function of the output
†	Pseudo inverse matrix operator

INTRODUCTION

The basic idea in estimation theory is to obtain approximations of the true response by using information from a model and from any available measurements. The mathematical structure used to perform estimation is known as an observer. The optimal observer for linear systems subjected to broad band disturbances is the Kalman Filter (KF). In the classical presentation of the filter the gain, K, is computed given the model parameters and the covariance of the process and the measurement noise, Q and R, respectively. Since Q and R are seldom known a priori work to determine how to estimate these matrices from the measured data began soon after introduction of the filter. An important result, apparently first published by Son and Anderson [1], is the fact that the information needed to determine K is encoded in the output signals and thus, given sufficiently long data sequences K can be estimated from the measurements without the need to know or estimate the Q, R matrices. Perhaps the most widely quoted strategies to carry out the estimation of K are due to Mehra [2] who built on the work by Heffes [3] and the subsequent paper by Carew and Bellanger [4], both techniques being iterative in nature. A noteworthy contribution from this early work is the contributions by Neethling and Young [5], who suggested some computational adjustments that could be used to improve accuracy.

The problem of computing K can be approached, of course, as one of estimating Q and R because once these matrices are known K follows from well known relations. If all that is of interest is K, therefore, one can take the direct approach, i.e., from the data to K, or the indirect one that estimates Q and R first. Nevertheless, if in addition to estimating the state one is interested in the covariance of the state error then the computation of Q and R becomes necessary since there is no direct approach (at least not known to the writers) to go from the data to the state error covariance P. The two approaches that have received most attention in the Q, R estimation problem are based on: 1) correlations of the innovation sequence and 2) correlations of the output.

In the innovations approach one begins by “guessing” a filter gain (usually be estimating some Q and R from whatever a priori knowledge there may be) and then the approach obtains the optimal

gain, the K, from analysis of the resulting innovations. In the limiting case where the output sequences are very long (infinite in principle) the results converge to the exact solution independently of the initial "guess". In the real situation, however, where one must operates with finite length sequences (finite precision and so on) the results for Q and R do depend on the initial estimation and a question that arises is whether the answer is sensitive or not. . In this regard, Mehra [6] suggested that the estimation of Q and R could be repeated by starting with the gain obtained in the first attempt but this expectation was challenged by Carew and Bellanger [4], who noted that the exact gain is used as the initial guess the approximations in the approach are such that the correctness will not be confirmed. Some observations on the issue of the initial gain are presented later in this paper.

The output correlation approach can be viewed as a special case of the innovations where the initial gain is taken as zero. In practice, however, the output correlation alternative has been discussed as a separate approach (probably) because the mathematical arrangements that are possible in this special case cannot be easily extended to the case where K ≠ 0. It is opportune to note that some recent work on the estimation of Q and R has taken place in the structural health monitoring community but the work appears to have been done in isolation of the classical works noted previously, [7]. Part of the motivation for this paper is to offer a concise review of the classical correlation based approaches that may prove useful in the structural engineering community. The paper is organized as follows: the next section provides a brief summary of the KF particularized to a time invariant linear system with stationary disturbances (which is a condition we have implicitly assumed throughout the previous discussion). The following two sections review the formulations to estimate Q and R using the innovations and output correlations and the paper concludes with a numerical example.

THE KALMAN FILTER

Consider a time invariant linear system with unmeasured disturbances w(t) and available measurements y(t) that are linearly related to the state vector x(t). The system has the following description in sampled time

$$x_{k+1} = Ax_k + Bw_k \tag{1}$$

$$y_k = Cx_k + v_k \tag{2}$$

where $A \in \mathbb{R}^{nxn}$, $B \in \mathbb{R}^{nxr}$ and $C \in \mathbb{R}^{mxn}$ are the transition, input to state and state to output matrices, $y_k \in \mathbb{R}^{mx1}$ is the measurement vector and $x_k \in \mathbb{R}^{nx1}$ is the state. The sequence $w_k \in \mathbb{R}^{nx1}$ is known as the process noise and $v_k \in \mathbb{R}^{mx1}$ is the measurement noise. In the treatment here, it is assumed that these are uncorrelated Gaussian stationary white noise sequences with zero mean and covariance of Q and R, namely

$$E(w_k) = 0 \quad E(w_k w_j^T) = Q\delta_{kj} \tag{3a,b}$$

$$E(v_k) = 0 \quad E(v_k v_j^T) = R\delta_{kj} \tag{4a,b}$$

$$E(v_k w_j^T) = 0 \tag{5}$$

where δ_{kj} denotes the Kronecker delta function and $\mathrm{E}(\cdot)$ denotes expectation. For the system in eqs.1 and 2 the KF estimate of the state can be computed from

$$\hat{x}_{k+1}^{-} = A\hat{x}_k^{+} \tag{6}$$

$$\hat{x}_k^{+} = \hat{x}_k^{-} + K(y_k - C\hat{x}_k^{-}) \tag{7}$$

where $\hat{x}_k^{+}$ is the estimate after the information from the measurement at time k is taken into consideration and $\hat{x}_k^{-}$ is the estimate before. The (steady state) Kalman gain K can be expressed in a number of alternative ways, a popular one is [8]

$$K = PC^T(CPC^T + R)^{-1} \tag{8}$$

where P, the steady state covariance of the state error, is the solution of the Riccati equation

$$P = A(P - PC^T(CPC^T + R)^{-1}CP)A^T + BQB^T \tag{9}$$

The KF provides an estimate of the state for which P is minimal. The filter is initialized as follows

$$\hat{x}_0^{+} = E[x_0] \tag{10}$$

$$P_0 = E[(x_0 - \hat{x}_0)(x_0 - \hat{x}_0)^T] \tag{11}$$

INNOVATIONS CORRELATION APPROACH FOR Q AND R ESTIMATION

We begin with the expression for the covariance function of the innovation process for any stable observer with gain K_0. As shown by Mehra [2] this function is

$$l_j = CPC^T + R \qquad j = 0 \tag{12}$$

$$l_j = C[A(I - K_0C)]^{j-1}A[PC^T - K_0(CPC^T + R)] \qquad j > 0 \tag{13}$$

where the covariance of the state error in the steady state, as shown by Heffes [3] is the solution of the Riccati equation

$$P = A(I - K_0C)P(I - K_0C)^T A^T + AK_0RK_0^T A^T + BQB^T \tag{14}$$

Estimation of PC^T

Listing explicitly the covariance function for lags one to ℓ one has

$$l_1 = CA(PC^T - K_0l_0) \tag{15}$$

$$l_2 = C[A(I - K_0C)]A(PC^T - K_0l_0) \tag{16}$$

$$l_3 = C[A(I - K_0C)]^2A(PC^T - K_0l_0) \tag{17}$$

.....

$$l_\ell = C[A(I - K_0C)]^{\ell-1}A(PC^T - K_0l_0) \tag{18}$$

from where one can write

$$L = Z(PC^T - K_0l_0) \tag{19}$$

In which,

$$L = \begin{bmatrix} l_1 \\ l_2 \\ l_3 \\ \dots \\ l_\ell \end{bmatrix} \qquad Z = \begin{bmatrix} CA \\ C[A(I - K_0 C)]A \\ C[A(I - K_0 C)]^2 A \\ \dots\dots \\ C[A(I - K_0 C)]^{\ell-1} A \end{bmatrix} \tag{20a,b}$$

As can be seen, matrix Z is the observability block of an observer whose gain is K_0 post-multiplied by the transition matrix A. On the assumption that the closed loop is stable and observable, one concludes that Z attains full column rank when ℓ is no larger than the order of the system, *n*. Accepting that the matrix is full rank one finds that the unique least square solution to eq.19 is

$$PC^T = K_0 l_0 + Z^\dagger L \tag{21}$$

where $Z^\dagger$ is the pseudo-inverse of Z. On the premise that the model is known without error (which is a strong assumption), error in solving eq.21 for PC^T results from the fact that the innovations covariance is approximate due to the finite duration of the signals.

Estimation of R

Having obtained an estimate for PC^T, the covariance of the output noise R can be estimated from eq.12 as

$$\hat{R} = \hat{l}_0 - C(\hat{P}\hat{C}^T) \tag{22}$$

where the hats are added to emphasize that the quantities are estimates.

Estimation of Q

Derivation of an expression to estimate Q is significantly more involved than the one for R. The process begins by replacing, in eq.14, the covariance R by its expression in terms of the autocorrelation at zero lag. After some algebra one gets

$$P = APA^T + M + BQB^T \tag{23}$$

where

$$M = A[-K_0 CP - PC^T K_0{}^T + K_0 l_0 K_0{}^T]A^T \tag{24}$$

Now consider a recursive solution for P in eq.23. In a first substitution one has

$$P = A(APA^T + M + BQB^T)A^T + M + BQB^T \tag{25}$$

or

$$P = A^2 P(A^2)^T + AMA^T + ABQB^T A^T + M + BQB^T \tag{26}$$

and after *q* substitutions one gets

$$P = A^q P (A^q)^T + \sum_{j=0}^{q-1} A^j M (A^j)^T + \sum_{j=0}^{q-1} A^j BQB^T (A^j)^T \tag{27}$$

Before solving eq.27 for Q it is first necessary to extract the set of equations for which only knowledge of PC^T (estimated by eq.21) is necessary. These equations are attained by post-multiplying both sides of eq.27 by C^T and pre-multiplying by CA^{-q}. This gives

$$CA^{-q} PC^T = CP(A^q)^T C^T + CA^{-q} \sum_{j=0}^{q-1} A^j M (A^j)^T C^T + CA^{-q} \sum_{j=0}^{q-1} A^j BQB^T (A^j)^T C^T \tag{28}$$

Given that P is symmetric, the CP product to the right of the equal sign can be expressed as $(PC^T)^T$, and one has

$$\sum_{j=0}^{q-1} CA^{j-q} BQB^T (A^j)^T C^T = CA^{-q} PC^T - (PC^T)^T (A^q)^T C^T - \sum_{j=0}^{q-1} CA^{j-q} M (A^j)^T C^T \tag{29}$$

For convenience we transpose both sides of eq.29 and get

$$\sum_{j=0}^{q-1} CA^j BQB^T (A^{j-q})^T C^T = (PC^T)^T (A^{-q})^T C^T - CA^q PC^T - \sum_{j=0}^{q-1} CA^j M (A^{j-q})^T C^T \tag{30}$$

where we've accounted for the fact that M is symmetrical. To shorten eq.30 we define

$$F_j = CA^j B \tag{31}$$

$$G_j = B^T (A^{j-q})^T C^T \tag{32}$$

and

$$s_q = (PC^T)^T (A^{-q})^T C^T - CA^q PC^T - \sum_{j=0}^{q-1} CA^j M \ \ (A^{j-q})^T C^T \tag{33}$$

So eq.30 becomes

$$\sum_{j=0}^{q-1} F_j Q G_j = s_q \tag{34}$$

Applying the *vec* operator to both sides of eq.34 one has

$$\sum_{j=0}^{q-1} (G_j^T \otimes F_j) \cdot vec(Q) = vec(s_q) \tag{35}$$

where $\otimes$ denotes the Kronecker product. Eq.35 can be evaluated for as many *q* values as one desires, although it is evident that all the equations obtained are not necessarily independent. Selecting *q* from one to *p* gives

$$H \cdot vec(Q) = S \tag{36}$$

where

$$h_q = \sum_{j=0}^{q-1} (G_j^T \otimes F_j) \quad H = \begin{bmatrix} h_1 \\ h_2 \\ .. \\ h_p \end{bmatrix} \quad S = \begin{bmatrix} vec(s_1) \\ vec(s_2) \\ .. \\ vec(s_p) \end{bmatrix} \tag{37a,b,c}$$

Eq.36 is the expression used in the innovations approach to obtain an estimate of Q. From its inspection one finds that H has dimensions m^2 p x r^2, where we recall that *m* and *r* represent the numbers of outputs and independent disturbances, respectively.

Structure in Q

The matrix Q is symmetrical and can often be assumed diagonal. The constraints of symmetry and/or a diagonal nature of Q can be reflected in a linear transformation of the form

$$vec(Q) = T \cdot vec(Q_r) \tag{38}$$

where *vec*(Q_r) is the vector of unknown entries in Q after all the constraints are imposed. In the most general case, where only symmetry is imposed, the dimension of T is $\mathrm{n}^2 \times \mathrm{r}(\mathrm{r}+1)/2$. However, since structural engineering problems are such that the largest possible value of r is n/2, T has a maximum dimension of $\mathrm{n}^2 \times \mathrm{n}(\mathrm{n}+1)/4$. In the most general case in structures, therefore, the matrix H, after symmetry is imposed, is $\mathrm{m}^2\mathrm{n} \times \mathrm{n}(\mathrm{n}+1)/4$. A necessary (albeit not sufficient) condition for a unique solution for Q, therefore, is $\mathrm{m} \geq \sqrt{\mathrm{n}+1}/2$.

Enforcing Positive Semi-Definitiveness

By definition, the true matrix Q is positive semi-definite (i.e., all its eigenvalues are ≥0), however, due to approximations, the least square solution may not satisfy this requirement. In the general case one can satisfy positive semi-definitiveness by recasting the problem as an optimization with constraints. Namely, minimize the norm of $(H \times vec(Q) - S)$ subject to the constraint that all eigenvalues of Q ≥ 0. This problem is particularly simple for our case, where Q is diagonal. In this scenario the entries in *vec*(Q) are the eigenvalues and all that is required is to ensure that they are not negative. Also, here is a suitable application for the always converging non-negative least squares solution [9]. This algorithm is used for the numerical example provided at the end of this paper.

OUTPUT CORRELATION APPROACH FOR Q AND R ESTIMATION

As in the previous case, it is assumed that the state is stationary and that the process and measurement noise are white and uncorrelated with each other, namely

$$E(x_{k+1} x_{k+1}^T) = \Sigma \tag{39}$$

$$E(x_k v_k^T) = 0 \tag{40}$$

$$E(x_k w_k^T) = 0 \tag{41}$$

$$E(w_k v_k^T) = 0 \tag{42}$$

Estimation of R

The covariance function of the output is

$$\Lambda_i = E(y_{k+i} y_k^T) \tag{43}$$

Substituting eqs.2 and enforcing the assumptions in 40-42 one gets

$$\Lambda_i = C(E\left(x_{k+i} x_k^T\right))C^T \tag{44}$$

From eq.1 one can show that

$$x_{k+i} x_k^T = A^i x_k x_k^T + A^{i-1} B w_k x_k^T + A^{i-2} B w_{k+1} x_k^T + \ldots + B w_{k+i-1} x_k^T \tag{45}$$

Taking expectations on both sides of eq.45 gives

$$E(x_{k+i} x_k^T) = A^i E(x_k x_k^T) \tag{46}$$

which when substituted into eq.44 gives

$$\Lambda_0 = C\Sigma C^T + R \quad for \quad i = 0 \tag{47}$$

$$\Lambda_i = C A^i \Sigma C^T \quad for \quad i \neq 0 \tag{48}$$

Defining

$$G = E(x_{k+1} y_k^T) \tag{49}$$

and substituting eqs.1 and 2, then imposing the assumptions in eqs.40-42 one gets

$$G = A\Sigma C^T \tag{50}$$

Writing out the covariance function in eq.48 for i=1,2…*p* one has

$$\begin{Bmatrix} \Lambda_1 \\ \Lambda_2 \\ . \\ \Lambda_p \end{Bmatrix} = \begin{bmatrix} C \\ CA \\ . \\ CA^{p-1} \end{bmatrix} A\Sigma C^T \tag{51}$$

or, substituting Eq.50 into Eq.51 and solving for G one has

$$G = Ob_p^{\dagger} \cdot \begin{Bmatrix} \Lambda_1 \\ \Lambda_2 \\ . \\ \Lambda_p \end{Bmatrix} \tag{52}$$

From eq.47 and eq.50 one gets

$$R = \Lambda_0 - CA^{-1}G \tag{53}$$

Estimation of Q

Substituting eq.1 into eq.39 and enforcing the operating assumptions gives

$$\Sigma = A\Sigma A^T + BQB^T \tag{54}$$

Applying the *vec* operator to eqs.54 and eq.50 one obtains

$$vec(\Sigma) = \left(I - (A \otimes A)\right)^{-1}(B \otimes B)vec(Q) \tag{55}$$

and

$$vec(G) = \left(C \otimes A\right)vec(\Sigma) \tag{56}$$

Combining eqs.55 and 56 gives

$$vec(G) = V \cdot vec(Q) \tag{57}$$

where

$$V = \left(C \otimes A\right)\left(I - (A \otimes A)\right)^{-1}(B \otimes B) \tag{58}$$

Structure in Q

The observations made in the case of the innovations approach apply here also.

Enforcing Positive Semi-Definitiveness

The observations made in the case of the innovations approach apply here also.

NUMERICAL EXAMINATION

We consider the 5-DOF spring-mass structure depicted in fig.1. The first un-damped frequency is 2.66Hz and the 5th is 16.30Hz. Damping is classical with 2% in each mode and the stochastic disturbance, having a Q = 1, is applied at the 1st coordinate. Velocity measurements are taken at the 3rd coordinate and the exact response is computed at 50Hz sampling. Noise in the output is such that R = 5x10^{-4}.

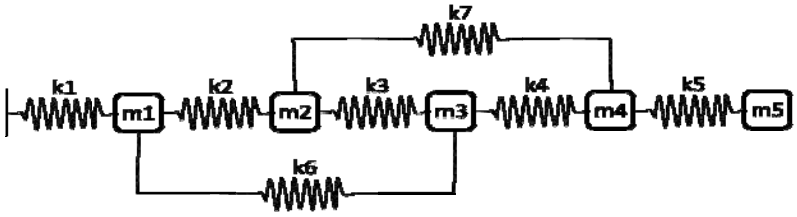


Fig.1 5-DOF spring-mass structure (m_1-m_5 = 0.05; k_1,k_3,k_5,k_7 =100; k_2,k_4,k_6=120)

Two hundred simulations are performed on each of four different cases to investigate the affect of duration and number of lags on the estimation of the steady state error covariance matrix P. The cases are defined as follows: Case I: duration 200sec lags=40; Case II: duration 200sec. lags=10; Case III: duration 20sec. lags=40; and Case IV: duration 20sec. lags=10. Estimates of Q and R are obtained from the presented methods and substituted into eq.9 for the calculation of P. On the basis that eq.9 requires Q and R to be positive definite, all negative estimations of R are replaced by zero. Results are presented for the diagonal elements of the covariance of the state error as histograms in figs. 2 and 3; m and s represent the mean and standard deviation of the estimate and t represents the true value. We note the expected value of the maximum displacement response in the first degree of freedom is 0.07 and %10 error represents 4.9×10^{-5} variance.

For conciseness only cases I and II are presented in figs. 2 and 3. These cases represent the most accurate estimations for both approaches. An interesting observation is that case I is most accurate for the innovations approach (many lags) while II (few lags) is most accurate for the output approach. One notes that both approaches have means in relatively good agreement with the true values but comparison of standard deviations show that the innovations approach is the more consistent estimator. As expected, the results for the shorter duration signals deteriorated notably (not shown for brevity).

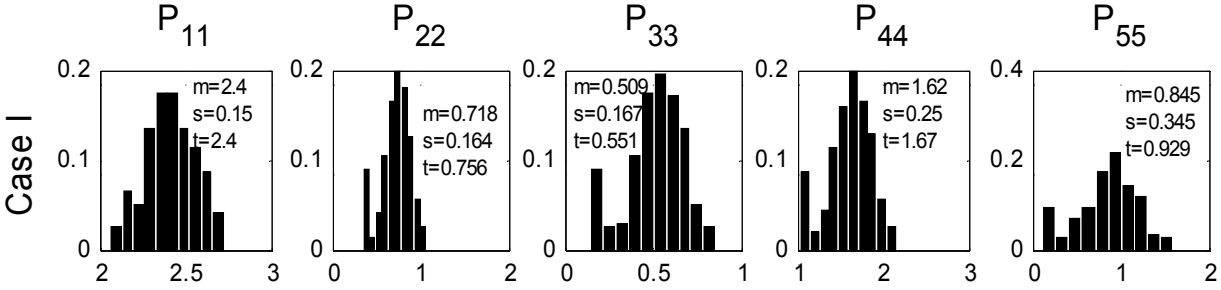


Fig.2 Histograms of the first five diagonal elements of the state error covariance obtained using estimates of Q and R from the innovations approach.

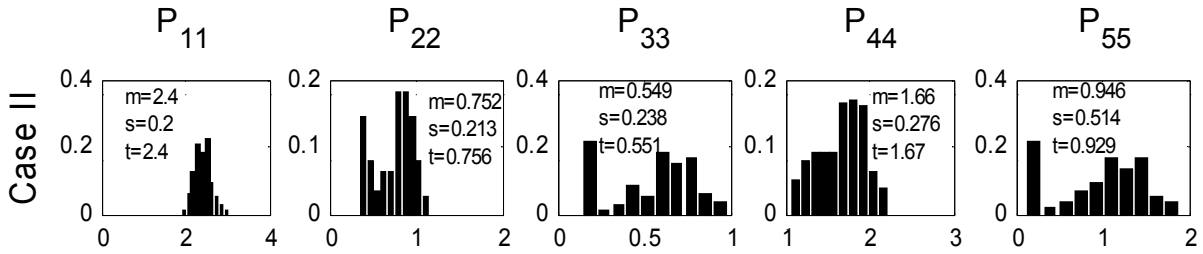


Fig.3 Histograms of the first five diagonal elements of the state error covariance obtained using estimates of Q and R from the output approach.

Influence of the Initial Gain

Intuitively one expects that the closer the initial guess is to the true Kalman gain the more accurate the results will be. This is generally true but exceptions are possible and tend to occur if some poles of the closed loop observer are small enough to become negligible for powers less than the system order because in this case some of the singular values in Z (eq.20b) can become very small and can lead to numerical trouble. To exemplify the previous observations, we considered the same system illustrated

in fig.1, except that output measurements are assumed available not only at coordinate 3 but also at 5. The covariance matrices are Q=1, $R=5*10^{-7}I$. Fig.4 shows histograms of the estimated Q based on 200 simulations for the case where the initial gains are: a) zero b) the exact KF and c) a filter based on Q = 1 R = 3I. The duration of the signals is 40 secs and 10 lags are used. As can be seen, the poorest result is obtained when the initial gain is the exact Kalman gain.

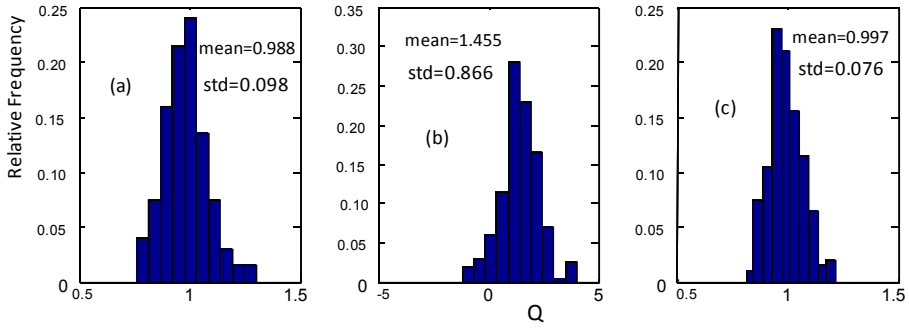


Fig.4 Histograms of Q with measurement noise $R = 5e^{-7}$ x I: a) gain = 0, b) gain = exact KF and c) gain = KF for Q = 1, R = 3 x I

CONCLUDING COMMENTS

The classical innovations and output covariance techniques to estimate Q and R from output measurements were reviewed. Both methods lead to the solution of a linear system of equations that is rather poorly conditioned so accuracy demands the use of long duration signals to minimize error in the correlation function estimates. In the numerical example signals with duration on the order of 40 times the period of the slowest mode proved inadequate. When the duration is 400 times the fundamental period the mean of 200 simulations proved in good agreement with the covariance of the state error but even then the coefficients of variation are not small. Specifically, in the innovations approach, which proved most accurate, the coefficient of variation of the 5th entry in the covariance of the state error is 0.4. Numerical results suggest that accuracy is promoted by having an initial gain that is close to the Kalman gain but care must be exercised to avoid initial gains that exacerbate the poor conditioning of the coefficient matrix that must be pseudo-inverted. While it is true that a second cycle where the “guess” for K is taken as the value from the first cycle is not guaranteed to provide improvement, this strategy, as suggested by Mehra [6] is expected to improve accuracy when the initial gain is far from the mark.

REFERENCES

[1] Son, L.H. and B.D.O. Anderson, “Design of Kalman Filters Using Signal Model Output Statistics,” Proc.IEE, Vol.120, No.2, February 1973, pp.312-318

[2] Mehra, R. K. “On the identification of variance and adaptive Kalman filtering”, IEEE Transactions on Automatic Control, Vol. 15, No.2, 1970, pp. 175-184.

[3] H. Heffes, “The effect of erroneous models on the Kalman filter response,” IEEE Transactions on Automatic Control, AC-11, July, 1966, pp. 541-543.

[4] Carew, B. and Belanger, P.R., “Identification of Optimum Filter Steady-State Gain for Systems with Unknown Noise Covariances”, IEEE Transactions on Automatic Control, Vol.18, No.6, 1974, pp. 582–587.

[5] C. Neethling and P. Young. “Comments on identification of optimum filter steady-state gain for systems with unknown noise covariances”, IEEE Transactions on Automatic Control, Vol.19, No.5, 1974, pp. 623-625.

[6] Mehra, R.K. “Approaches to adaptive filtering,” IEEE Transactions on Automatic Control, Vol. AC-17, Oct., 1972, pp. 693-698.

[7] Yuen, K.V., Hoi, K.I., and Mok, K.M., “Selection of Noise Parameters for Kalman Filter”, Journal of Earthquake Engineering and Engineering Vibration (Springer Verlag), Vol.6, No.1, 2007, pp. 49-56.

[8] Dan Simon, “Optimal State Estimation: Kalman, H Infinity, and Nonlinear Approaches”, Wiley-Interscience, 2006

[9] Lawson, C.L., & Hanson, R.J. "Solving Least Squares Problems", Prentice-Hall, 1974, Chapter 23, p.161.